

SAVIDU HERATH

Software Engineering Undergraduate | Full-Stack Developer (MERN Stack)

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PROFESSIONAL SUMMARY

Results-driven Software Engineering undergraduate at SLIIT (GPA: 3.04) with proven experience designing and delivering production-grade web, mobile, and desktop applications across JavaScript (MERN), Java (Spring Boot), and Go ecosystems. Demonstrated expertise in architecting scalable microservices, offline-first sync engines, and resilient event-driven systems using RabbitMQ, Redis, and multi-database architectures (MongoDB, PostgreSQL, SQLite). Hands-on experience containerizing services with Docker, orchestrating deployments with Kubernetes, provisioning AWS cloud infrastructure, and ensuring software quality via Cypress end-to-end testing and Agile/Scrum practices. Eager to contribute to dynamic engineering teams as a Software Engineering Intern, bringing strong technical fundamentals, a collaborative mindset, and a commitment to continuous learning.

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), Go (Golang), Java

Frameworks & Libraries: React.js, React Native, Angular, Node.js, Express.js, Spring Boot, Electron

Databases: MongoDB, PostgreSQL, SQLite, Redis, MySQL

Web Technologies: HTML5, CSS3, RESTful APIs, AMQP (Message Queuing), DOM Manipulation

DevOps & Cloud: Docker, Docker Compose, Kubernetes, AWS (EC2, S3), Vercel, Render, CI/CD Pipelines

Testing: Cypress (End-to-End Testing)

Tools & Platforms: Git, GitHub, RabbitMQ, Postman, VS Code, IntelliJ IDEA

Concepts: Microservices Architecture, Event-Driven Architecture, Offline-First Systems, Containerization, Role-Based Access Control (RBAC), API Gateway Pattern, JWT Authentication, Agile/Scrum, Data Structures & Algorithms, Responsive Design

PROJECTS & PRACTICAL EXPERIENCE

[Pharmacy Management System](#) | *Java · Spring Boot · React · MySQL · JWT · Spring Security*

- Developed a full-stack pharmacy management system using Java, Spring Boot, React, and MySQL, implementing modular backend services for medicine inventory, user management, supplier management, and role-based pharmacy operations.
- Implemented secure authentication and authorization using Spring Security and JWT, enforcing role-based access control for ADMIN, PHARMACIST, and CUSTOMER users across protected RESTful API endpoints.
- Designed and developed RESTful APIs using Spring Boot and Spring Data JPA, implementing CRUD operations, server-side pagination, search, filtering, stock-level monitoring, expiry tracking, and medicine management workflows.
- Built a responsive React frontend with protected routing and role-specific dashboards, providing dedicated interfaces for administrators and pharmacists to manage users, medicines, inventory, and operational data.
- Implemented an admin user management module with staff creation, role management, deletion controls, validation, and authorization safeguards to prevent unauthorized administrative actions.

[EmpowerHer – Community & Mentorship Platform](#) | *MongoDB · Express.js · React · Node.js · Vercel · Render · Cypress*

- Designed and developed a full-stack mentorship and community platform that connects mentors and mentees through structured matching, enabling profile creation, session scheduling, and community forum interactions across user roles.
- Built a scalable Node.js/Express.js backend with clearly defined RESTful API endpoints for user management, forum posts, mentorship session handling, and notification triggers, with MongoDB as the primary data store.

- Deployed the React frontend to Vercel and the backend API to Render, establishing a CI/CD-ready pipeline with environment-based configuration for development, staging, and production parity.
- Implemented JWT-based secure user authentication, granular role-based access control (mentee, mentor, admin), and responsive UI components to deliver an accessible and inclusive experience across all device sizes.
- Authored end-to-end test suites in Cypress covering critical user flows — including registration, login, mentor matching, and session booking — to validate functional correctness and prevent regressions in core features.

Legacy Lens — Local Language & Culture Preservation Platform | *React Native · Angular · Spring Boot · PostgreSQL*

- Developed a full-stack cultural heritage preservation platform connecting elders with youth creators through a content marketplace, gamified learning system, and searchable archive, addressing digital-literacy barriers with a voice-guided, low-literacy-friendly interface..
- Engineered the Learning Engine module in React Native, including gamified quiz-based lessons, vocabulary flashcards with elder-recorded audio playback, and progress tracking with streaks and completion certificates, consuming RESTful APIs built with Spring Boot. Designed modular, reusable React components with consistent theming and responsive layouts, ensuring a seamless ordering experience across desktop and mobile devices.
- Designed reusable React Native components with a consistent, culturally-themed UI and structured navigation flow, ensuring an engaging and accessible learning experience for youth learners across the mobile app.
- Structured Spring Boot service and controller layers with PostgreSQL-backed entities for course tracks, lessons, and learner progress, applying clean separation between API, business logic, and data access layers; collaborated across a 4-member Agile Scrum team using rotating Product Owner/Scrum Master roles over 5 sprints.

Offline-First Inventory & Live Sync System | *Golang · React · Electron · SQLite · PostgreSQL · RabbitMQ · Redis · Docker*

- **Architected an offline-first distributed inventory system** using a dual-database architecture—leveraging local SQLite for zero-latency operations during network outages and central PostgreSQL for unified cloud data consolidation.
- **Engineered a resilient event-driven sync engine** in Go using RabbitMQ (AMQP), implementing transactional message queuing, bi-directional push/pull pipelines, and automatic retry mechanisms to ensure eventual data consistency.
- **Containerized core services and data stores** (PostgreSQL, RabbitMQ, Redis) using Docker Compose, configuring custom bridge networking and host port mappings to enable reliable inter-service communication across local environments.
- **Integrated Redis caching** for high-frequency dashboard analytics and implemented strategic cache invalidation triggers upon database sync events to drastically reduce heavy read queries.
- **Developed a cross-platform desktop UI** using React and Electron, featuring real-time sync telemetry, status indicators (Live Sync / Sync Pending), and robust local state fallback during offline scenarios.

EDUCATION

BSc (Hons) Software Engineering — SLIT, Malabe, Sri Lanka

2022 – Present

Currently in 3rd Year | GPA: 3.04 | Relevant Coursework: Data Structures & Algorithms, Software Architecture, Web Technologies, Database Management, Cloud Computing

Diploma in English — Professional Communication

2023

Developed professional written and verbal communication skills for technical environments, enhancing team collaboration and documentation quality.

GCE Advanced Level (Physical Science Stream) — Kekirawa Central College

2022

GCE Ordinary Level — Kekirawa Central College

2019

ADDITIONAL INFORMATION

Soft Skills: Teamwork · Problem-Solving · Communication · Adaptability · Responsibility · Positive Attitude

Languages: English (Professional Proficiency) | Sinhala (Native)